

A topology-derived flow inference approach to optimize sensor placement for effective inflow and infiltration detection in sewer networks

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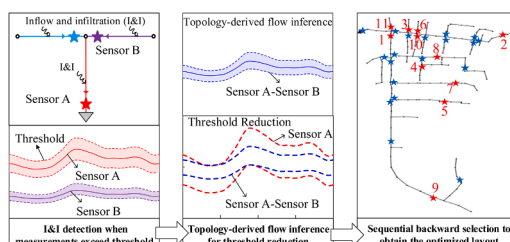
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HIGHLIGHTS

- A novel topology-derived flow inference method is proposed for enhancing inflow and infiltration detection in sewer networks.
- A new Detection Threshold metric is formulated to optimize sensor placement for superior sensitivity to minor I&I events.
- The deterministic Sequential Backward Selection algorithm efficiently yields near-optimal, interpretable sensor layouts.
- Sensor density in high building-density areas is prioritized, significantly boosting overall detection performance.
- The framework demonstrates robust scalability and outperforms benchmark methods in both synthetic and large-scale real-world networks.

GRAPHICAL ABSTRACT



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ABSTRACT

Inflow and infiltration (I&I) in urban sewer networks increase both the delivery loads and overflow risks, thereby compromising environmental safety. While sensor-based detection of I&I is promising, a key limitation in most current applications is their reliance on fixed, isolated detection thresholds derived from individual sensors. This approach prevents the synthesis of information from multiple sensors and inherently limits the detectability of I&I events. To bridge this gap, this study introduces a novel topology-derived flow inference (TDFI) method that quantifies the pipe-specific minimum detectable I&I flow by synthesizing data from upstream sensors in sewer

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networks. Based on this method, a new detection threshold metric is formulated for sensor placement strategy (SPS) optimization, accompanied by an efficient Sequential Backward Selection (SBS) approach that deterministically constructs hierarchical sensor subsets through an iterative removal of the least-contributing sensors. Evaluation across three case studies (one synthetic and two real-life) demonstrates that the TDFI-based optimization framework identifies robust SPSs, leading to significantly improved overall detection performance for minor I&I events. A key finding is that prioritizing sensor density in high building-density areas significantly enhances overall detection performance. Comparative analysis shows that SBS achieved performance close to that of the DE-based optimizer while providing deterministic layouts under various budgets, offering a computationally efficient complementary approach. The core contribution of this study is to propose a topology-informed framework for SPS optimization in support of I&I detection, integrating upstream flow inference to maximize the detection capability of sensor layouts.

1. Introduction

Urban sewer networks are critical infrastructure designed to convey wastewater to treatment facilities, safeguarding public health and the environment (Jia et al., 2021a). However, their operational efficacy is frequently compromised by inflow and infiltration (I&I), a phenomenon in which extraneous water enters the system illicitly (Zeydinejad et al., 2024). This intrusion results in a heightened delivery load, a diminished conveyance capacity, and an elevated risk of sewer overflows (Li et al., 2023). Consequently, the accurate and timely detection of I&I events has become a primary objective in sewer system management (Ge et al., 2024). Conventional approaches to detecting I&I generally fall into two categories. The first relies on macro-scale estimation methods, such as flow analysis or pollutant mass balance. The second involves localized physical inspections, including smoke testing and closed-circuit television surveys. However, macro-scale techniques often lack the spatial resolution and hence cannot pinpoint specific entry points. In contrast, while physical inspections can accurately locate defects, they are costly, labor-intensive, and therefore impractical for large-scale assessments (Tatiparthi et al., 2025).

The advancement of Internet of Things (IoT) technologies has enabled the widespread deployment of sensors for I&I detection (Jia et al., 2021b). Typically, these sensors collect high-resolution hydraulic data (e.g., flow rate or water level), enabling I&I events to be identified through data-driven techniques more efficiently than with traditional inspections. However, the high costs of sensor procurement, installation, and maintenance make comprehensive network coverage economically unfeasible. Consequently, maximizing the effectiveness of a sensor placement strategy (SPS) under a limited budget becomes a critical task. Methodologically, SPS optimization involves two key elements: (i) defining an evaluation metric that quantitatively represents the detection capability (i.e., sensitivity to minor I&I) of a given layout, and (ii) effective and efficient solving of the resulting optimization problem. The first element requires a robust and physically meaningful quantifier of detection performance, whereas the second requires searching a high-dimensional sensor-location subset space.

Given that flow is the most direct indicator of I&I, this study focuses on optimizing the placement of flow sensors to address these dual objectives. Although water-level sensing is more common due to its lower cost, flow monitoring has been increasingly adopted in I&I assessment programs. As an example of engineering practice, in the Erhai Lake region (Yunnan, China), over 100 flow sensors have been deployed for I&I assessment, and parts of the monitoring layout have been iteratively repositioned due to the lack of a rigorous placement strategy. Therefore, focusing on flow-sensor placement for I&I detection is well motivated and practically relevant.

More specifically, a traditional data-driven method for I&I detection analyzes historical flow data to establish legitimate variation ranges, or detection thresholds (DTs), triggering alarms when new data exceed them. While effective for coarse identification, this conventional approach suffers from a critical shortcoming: it relies on static thresholds (e.g., a fixed 15 % deviation), which are calculated in isolation for individual sensors (Wu et al., 2025). This approach fails to synthesize

information from multiple, correlated flow sensors, which inherently limits sensitivity to minor I&I events. Crucially, sewer networks often have a tree-like topology, creating deterministic mass balance relationships (e.g., predicting downstream flow from upstream sums or isolating branch flows by difference). By synthesizing data according to this topological constraint, it is possible to derive more refined DTs, thereby enhancing detection sensitivity for minor I&I events. However, despite its potential, the integration of topology-derived flow inference to synthesize sensor data for threshold refinement remains unexplored.

Furthermore, the optimization of sensor layout is guided by performance evaluation metrics, as they define how detection capability is quantified. Existing metrics are broadly categorized into two types. The first type is information theoretic metrics, which aim to maximize the information gain of the sensor network. Representative examples include entropy-based methods introduced by Krstanovic and Singh (1992) and extended by researchers such as Alfonso et al. (2010), Lee (2013) and Yazdi (2018). The second type is event detection reliability metrics, which evaluate system performance based on detection rates against predefined event scenarios and thresholds (Ostfeld and Salomons 2004; Banik et al. 2017). However, a key limitation of information theoretic metrics is that they often function as data-driven "black-box" tools, prioritizing statistical patterns while neglecting the physical flow relationships inherent in sewer topology. Similarly, event detection reliability metrics depend on binary detection logic, making outcomes highly sensitive to assumed event parameters (Du et al. 2024). Consequently, neither approach systematically exploits the network's inherent topology to inform sensor placement, which can lead to suboptimal SPSs.

In addition, SPS optimization is a challenging discrete combinatorial problem. While stochastic evolutionary algorithms (EAs) are widely used and offer strong global search capability, their practical performance may depend on algorithmic settings and random initialization, often requiring case-specific adjustments. In this context, deterministic methods provide useful complementary approaches when rapid and reproducible solutions are required.

To bridge these gaps, this study proposes a topology-derived flow inference (TDFI) method to refine pipe-specific DTs by synthesizing information from upstream sensors in sewer networks. Based on the TDFI method, we formulate a corresponding DT-based evaluation framework and further develop a deterministic Sequential Backward Selection (SBS) algorithm to obtain reproducible near-optimal solutions to the resulting SPS optimization problem through iterative backward elimination. The effectiveness of the integrated TDFI-based framework is evaluated by comparing it with uniformity-maximized designs and Differential Evolution (DE) algorithms on both synthetic and real-world sewer networks. In this study, the framework is specifically developed and demonstrated for SPS optimization in support of I&I detection. The main contributions and novelties of this research are outlined as follows:

- (i) We propose a new topology-derived flow inference (TDFI) method that enhances the detection performance of sensor networks and overcomes the limitation of relying on fixed, isolated thresholds by synthesizing information according to topology-

derived flow relationships, thereby improving sensitivity to minor I&I events.

- (ii) We formulate a new DT-based metric that uses TDFI to quantify the minimum detectable flow for each pipe. Coupled with a DT-minimization optimization framework, this continuous metric enables more informative SPS assessment and design than traditional binary detection metrics.
- (iii) We develop a deterministic Sequential Backward Selection (SBS) method for SPS optimization through iterative removal of the least-contributing sensors. In addition, the comparative SPS analysis provides new insights into the effect of sensor density on overall detection capability.

2. Methodology

Fig. 1 illustrates the research objectives together with the overall framework of this study. To achieve the first objective, three evaluation metrics are introduced. As illustrated in the figure, Average Detection Threshold (ADT) is presented as a novel metric for SPS optimization. It is calculated by aggregating the Individual Pipe DTs (IPDTs) of all pipes, weighted by their lengths. Each IPDT reflects the minimum detectable I&I flow for a specific pipe and is derived from the DT, which reflects the base flow uncertainty associated with the sensors monitoring the pipes. Two benchmark metrics—Event Detection Reliability (EDR) and Distribution Uniformity (DU)—are employed for comparative evaluation.

For the second objective, the study systematically compares SPSs optimized using different metrics across three sewer networks: a simplified three-pipe network for controlled demonstration, and two real-world networks for practical evaluation. In the simple network, the advantages of the DT metric are demonstrated by highlighting the limitations of the traditional EDR metric. In the real-world cases, layout characteristics of ADT-minimized versus DU-maximized SPSs are compared to reveal their underlying mechanisms.

The third objective evaluates the proposed SBS method through performance comparisons with the DE algorithm, establishing SBS as a deterministic and computationally efficient alternative for SPS optimization. This framework enables systematic evaluation of sensor layouts under various budget constraints, supporting practical cost-benefit analysis for real-world engineering applications.

2.1. Topology-derived flow inference (TDFI) method and DT calculation

2.1.1. Concept of detection enhancement based on TDFI

Sewer networks typically feature a branching, tree-like topology. Unlike looped water distribution networks, this topological determinism enables bidirectional flow estimation: downstream totals can be predicted by aggregating known upstream flows; conversely, unknown

branch flows can be isolated by subtracting known upstream contributions from a downstream total. The Topology-derived Flow Inference (TDFI) method leverages this principle to synthesize sensor information and enhance anomaly detection sensitivity, as illustrated in Fig. 2. Specifically, Fig. 2a shows a network topology with two in-series sensors, B (upstream) and A (downstream). Their corresponding flow measurements and associated variation ranges are plotted in Fig. 2b. Within this framework, the DT is defined as the variation boundary, representing the minimum anomaly magnitude that can be distinguished from legitimate variations.

Given that Pipe P2 converges with P1 into Pipe P3, applying the TDFI method yields the differential signal ($A - B$). As illustrated in Fig. 2c, this differential signal captures the net inflow from the unmonitored branch and effectively isolates flow variations specific to the confluence section between the two sensors. Consequently, as shown in Fig. 2d, the inferred ($A - B$) signal exhibits a narrower variation range compared with the original Sensor A signal, leading to a improved detection sensitivity for localized anomalies. This narrowing effect arises from the mass conservation principle and topological constraints, which govern the relationship between upstream and downstream flows, thereby leading to more refined DTs and improving sensitivity for localized anomalies.

For illustration purposes, Fig. 3 provides a numerical example showing how the TDFI method refines DTs. In the single-sensor case (Fig. 3a), the IPDT for each pipe is calculated as 18 L/s. This value is derived by applying a 15 % variation tolerance (adopted from Wu et al., 2025) to the baseline flow of 120 L/s. Although the specific tolerance level may vary across different networks, it does not affect the relative ranking of SPSs as discussed later—since all IPDTs scale proportionally with the chosen tolerance percentage.

Fig. 3b demonstrates the dual effect of adding a sensor in Pipe 2: directly reducing its IPDT from 18 L/s to 6 L/s, and indirectly refining the IPDT for Pipes 1 and 3 to 13.42 L/s through the application of the TDFI method followed by DT refinement, as summarized below:

- (i) Application of the TDFI method: Based on mass balance at the junction where Pipes 1 and 2 join Pipe 3, a virtual sensor is defined to isolate the flow contribution from Pipe 1 and the upstream section of Pipe 3. Its flow is obtained from the differential measurement of the two physical sensors: $Q_{\text{virtual}} = Q_A - Q_B = 120 - 40 = 80$ L/s. Assuming a 15 % tolerance level, the inherent variation of this isolated component is $80 \times 0.15 = 12$ L/s.
- (ii) DT refinement based on the inferred flow component: The total flow in Pipe 3 is conceptually decomposed into two contributing components: the flow from the physically monitored Pipe 2 ($Q_B = 40$ L/s, variation = 6 L/s) and the flow from the branch monitored by the virtual sensor ($Q_{\text{virtual}} = 80$ L/s, variation = 12 L/s).

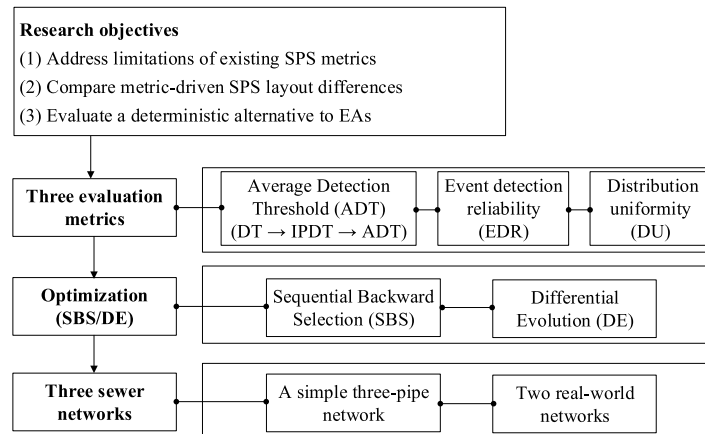


Fig. 1. Research objectives and overall framework of this study.

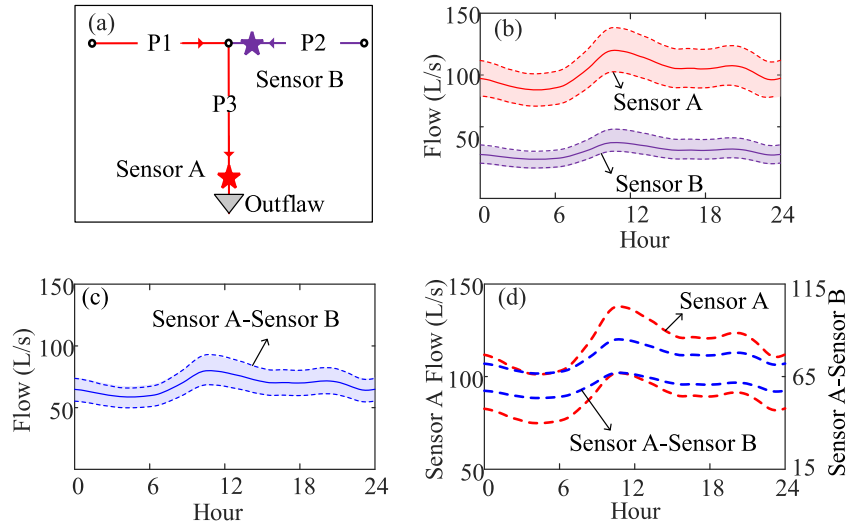


Fig. 2. Mechanism of detection enhancement using topology-derived flow inference.

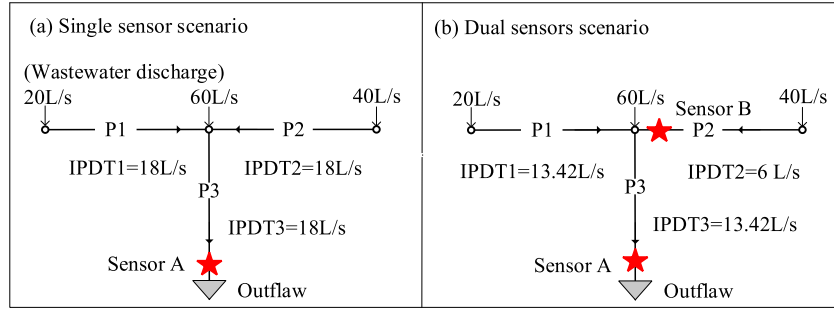


Fig. 3. Deriving a more refined IPDT using the TDFI method: a comparison between single- and dual-sensor scenarios.

s). Assuming statistical independence, the combined uncertainty is obtained using the Root-Sum-Square (RSS) method as $\sqrt{6^2 + 12^2} \approx 13.42 \text{ L/s}$. This approach avoids the over-estimation from simple arithmetic addition (Taylor and Kuyatt, 1994) and provides a more sensitive (13.42 vs. 18 L/s), source-attributable DT.

2.1.2. IPDT calculation for multi-sensor networks

Building on the TDFI method introduced above, the IPDT calculation for multi-sensor networks follows a three-step procedure based on sensor coverage and upstream uncertainty propagation.

Step 1. Determine the Net Flow Contribution of Each Sensor

Each sensor is assigned a set of pipes it monitors, based on network topology and sensor responsibilities. The net flow attributed to a sensor is calculated by summing the baseline flows of all pipes in its monitoring group:

$$Q_{\text{sensor}}^{(\text{net})} = \sum_{j \in \mathcal{S}_{\text{sensor}}} Q_j \quad (1)$$

where $\mathcal{S}_{\text{sensor}}$ is the set of pipes monitored by that sensor. This net flow represents the contribution from the pipe segments uniquely attributable to that sensor. For sensors with no upstream sensors, this net flow represents the total flow entering its monitored region.

Step 2. Establish Isolated DTs for Each Sensor

The isolated DT for a sensor is the uncertainty contributed by its own net flow, calculated by applying the variation tolerance α :

$$DT_{\text{sensor}}^{(\text{isolated})} = \alpha \cdot Q_{\text{sensor}}^{(\text{net})} \quad (2)$$

Where $DT_{\text{sensor}}^{(\text{isolated})}$ is the isolated DT for the sensor, $Q_{\text{sensor}}^{(\text{net})}$ is the net flow

attributed to the sensor (calculated in Step 1); α is the variation tolerance, set to 15 % in this study.

Step 3. Propagate Uncertainties to Obtain Pipe-Specific IPDTs

For a pipe group monitored by a sensor, its IPDT is the root-sum-square (RSS) combination of the sensor's isolated threshold and the isolated thresholds of all upstream sensors:

$$IPDT_{\text{group}} = \sqrt{(DT_{\text{sensor}}^{(\text{isolated})})^2 + \sum_{k \in \mathcal{U}_{\text{sensor}}} (DT_k^{(\text{isolated})})^2} \quad (3)$$

where $DT_k^{(\text{isolated})}$ represents the isolated DT for each upstream sensor k that contributes to the flow measured by the current sensor; $\mathcal{U}_{\text{sensor}}$ is the set of sensors upstream of the current sensor. Fig. 4 provides a numerical example for better understanding and practical application.

Specifically, the initial step identifies the pipes directly monitored by each sensor and sums their respective flow. As color-coded in Fig. 4, the configuration is as follows: Sensor A (purple) covers Pipe 1 with a flow of 10 L/s; Sensor B (red) observes Pipes 2 and 3, whose combined flow is 20 L/s; Sensor C (blue) is responsible for Pipes 4, 5, and 6, with a total flow of 40 L/s. Next, the isolated DT for each sensor is calculated, ignoring variation accumulation from other units. This is done by applying the assumed 15 % variation tolerance to each sensor's aggregated flow: the threshold for Sensor A (10 L/s) is 1.5 L/s; for Sensor B (20 L/s), 3 L/s; and for Sensor C (40 L/s), an isolated threshold of 6 L/s.

The final step computes the accumulative DT, which accounts for the propagation of variation from upstream sensors using the RSS method. Sensor A, with no upstream sensors, retains its threshold of 1.5L/s. For Sensor B, the threshold incorporates propagated variation from Sensor A, resulting in $\sqrt{1.5^2 + 3^2} = 3.35 \text{ L/s}$. For Sensor C, variations from both

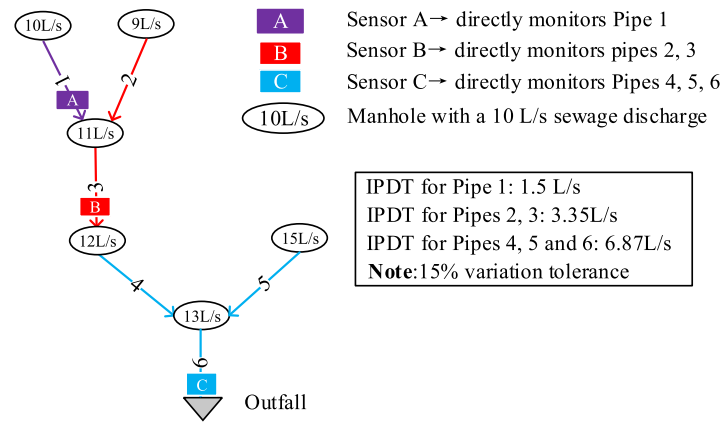


Fig. 4. Multi-sensor IPDT calculation in a complex sewer network.

Sensors (A and B) are propagated, resulting in a cumulative threshold of $\sqrt{1.5^2 + 3^2 + 6^2} = 6.87\text{L/s}$. Consequently, these cumulative thresholds are assigned to their respective pipe groups in Fig. 4: 1.5L/s for Pipe 1; 3.35L/s for Pipes 2 and 3; and 6.87L/s for Pipes 4, 5 and 6.

For unmonitored pipes (not shown in Fig. 4), the IPDT is set to the maximum potential I&I flow (i.e., the full-pipe capacity), determined using the Manning equation:

$$Q = \frac{\pi D^3 I^2}{43n} \quad (4)$$

where D is the pipe diameter, I is the slope, and n is the roughness coefficient. This value imposes a penalty for lack of monitoring coverage. The IPDT calculation for unmonitored pipes will be illustrated in Case Study 1.

2.1.3. Modeling assumptions and practical considerations

Note that three key assumptions underpin the proposed methodology, each associated with specific practical implications:

- (i) **Quasi-instantaneous mass conservation is assumed for topology-derived flow inference.** In real sewer networks, flow travel times introduce delays between upstream and downstream measurements. This issue can be mitigated by operating at appropriately long time scales (e.g., daily aggregated flows), where travel-time effects become negligible relative to the aggregation window. In practice, we propose using daily sewage discharge estimates derived from water billing data to establish the required mass conservation relationships, an approach supported by Zhang et al. (2021).
- (ii) **A uniform tolerance level (e.g., 15 %) is applied to different flow components.** In practice, the coefficient of variation (isolated DTs) typically decreases with increasing flow rate. However, because the DT remains dominated by the flow magnitude, this simplification preserves the relative ranking of DT values across the network. Consequently, the optimized SPS is robust to this assumption, with only minor positional adjustments possible among adjacent pipes of similar flow.
- (iii) **Statistical independence between flow components is assumed when combining uncertainties via RSS.** In practice, positive correlation ($\rho > 0$) may exist between flow components due to shared discharge patterns or common meteorological drivers. For two components, the generalized expression is: $\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2}$. While estimating correlation ρ would improve uncertainty propagation and ADT evaluation, network-wide correlations are difficult to quantify reliably because they

vary with catchment characteristics and the aggregation time scale. Therefore, we set $\rho = 0$ as a pragmatic baseline to keep the framework implementable and to focus on the core optimization methodology.

Finally, it should be emphasized that all three assumptions can be refined if site-specific data become available, without altering the core optimization framework of the proposed SBS method. Given these simplifying assumptions, the present uncertainty formulation should be interpreted as an engineering-oriented approximation, rather than as a fully field-validated statistical uncertainty propagation model.

2.2. Three SPS evaluation metrics

This section elaborates on three optimization metrics, each serving a distinct objective. The DT defines the system's sensitivity, where lowering the DT prioritizes the detection of minor I&I events. The Event Detection Reliability (EDR) quantifies the proportion of identifiable events, improving the EDR increases the number of captured incidents. The DU evaluates the spatial balance of sensor placement, enhancing the DU ensures equitable coverage for monitoring and localization.

2.2.1. DT metric

The DT is defined as the minimum anomaly magnitude that a SPS can reliably distinguish from legitimate variations. Originally proposed by Qi et al. (2018) to quantify the minimal detectable burst outflow in water distribution networks, the DT concept was subsequently adapted by Du et al. (2024) as a key metric for optimizing pressure sensor placement in these networks. In this study, we extend the application of DT-based metrics to sewer networks, motivated by the engineering principle that an optimal SPS should maximize sensitivity to anomalies.

Based on the IPDT calculation method detailed in Section 2.1, we introduce the Average DT (ADT) as a system-wide performance indicator. The ADT is defined as the pipe-length-weighted average of all IPDTs, as expressed in Eq. (1). This weighting approach accounts for the fact that longer pipes typically have a higher probability of infiltration, thus giving them greater influence on the overall system's detection performance.

$$\text{ADT} = \frac{\sum_{j=1}^m l_j \cdot (\text{IPDT}_j)}{\sum_{j=1}^m l_j} \quad (5)$$

where m represents the total number of pipes in the sewer network, l_j denotes the length of pipe j , IPDT_j is the Individual Pipe DT (IPDT) for pipe j .

2.2.2. Event Detection Reliability (EDR) metric

The Event Detection Reliability (EDR) metric quantifies the likeli-

hood that $An\ sps$ will successfully identify predefined abnormal events, typically measured as the ratio of detected events to the total number of simulation events. The mathematical formulation of EDR involves simulating a large number of potential events and calculating the proportion that is detected. For a given SPS, the EDR is computed as follows (Banik et al. 2017):

$$R = \frac{1}{S} \sum_{s=1}^S \delta_s \quad (6)$$

where S is the total number of simulated events, and δ_s is a binary indicator variable that equals 1 if the event is detected and 0 otherwise. In practical terms, if we simulate one event per pipe and assume each pipe has an equal probability of experiencing an event (e.g., uniform pipe length and infiltration), the EDR simplifies to the number of detectable pipes divided by the total number of pipes (see Case Study 1 in Section 4.1).

A key strength of the EDR metric lies in its intuitive and direct interpretation, aligning closely with the primary engineering objective of maximizing event detection probability. However, it is subject to notable limitations, particularly its inherent subjectivity in defining the intensity of simulated I&I events. This subjectivity can influence SPS optimization outcomes, as noted in Du et al. (2024) and further discussed in subsequent sections.

2.2.3. Distribution Uniformity (DU) metric

The Distribution Uniformity (DU) metric evaluates the spatial balance of a sensor placement design. It is calculated based on each sensor's direct monitoring responsibility, defined as the total length of pipes monitored by that sensor (see Fig. 4). While traditional metrics like joint entropy were used to maximize the information captured by a SPS, Wu et al. (2025) recognized that a high-information solution does not guarantee spatially balanced sensor deployment. To address this, they proposed DU as a crucial complementary criterion to the EDR, ensuring practical and equitable monitoring coverage. Specifically, DU is calculated as the inverse of the coefficient of variation of the pipe lengths monitored by each sensor, i.e. as follows:

$$DU = \frac{1}{C_v} = \frac{\bar{L}_s}{\sqrt{\frac{1}{NS} \sum_{s=1}^{NS} (L_s - \bar{L}_s)^2}} \quad (7)$$

Where L_s is the total length of pipes for which the sensor is the detector of I&I events, $\bar{L}_s$ is the average of these lengths across all sensors, and C_v is the coefficient of variation. Optimizing $A.\ sps$ to maximize DU ensures a balanced workload for pinpointing detected I&I event locations, preventing sensor overloading and underutilization.

2.3. SPS optimization

2.3.1. SPS optimization problem

The SPS optimization problem aims to identify the optimal subset of nodes within a sewer network, subject to a given budget constraint. It is a combinatorial problem in which B sensor locations must be selected from N candidate nodes to optimize one or more performance metrics for network monitoring, specifically the objective functions defined in Section 2.2. Decision variables are encoded as a vector $s = [s_1, s_2, \dots, s_B]$ of unique node IDs, where each $s_j \in \{1, 2, \dots, N\}$ corresponds to a sensor location that monitors the flow in the pipe immediately upstream. The sole explicit constraint is the sensor budget $|s| = B$, which limits the total number of deployed sensors. Given the combinatorial nature of the problem, heuristic or meta-heuristic optimization methods are required to obtain near-optimal solutions efficiently. The following subsections present two such approaches applied in this study.

2.3.2. Differential evolution based optimization

The DE algorithm is a population-based stochastic optimizer that

iteratively refines candidate solutions through mutation, crossover, and selection operations. Originally designed for continuous optimization, DE is adapted here to the discrete sensor placement problem by applying integer rounding after each mutation and crossover operation. This ensures that every candidate solution remains a valid integer vector representing a set of node IDs. In this study, each candidate SPS is encoded as $s = [s_1, s_2, \dots, s_B]$ where each s_j corresponds to a unique sensor location as defined in Section 2.3.1. The DE algorithm is applied to both ADT-minimization and DU-maximization problems, facilitating effective exploration of the combinatorial solution space. A detailed exposition of the DE algorithm and comprehensive performance comparisons with other evolutionary algorithms are beyond the scope of this paper; interested readers are referred to specialized literature (Zheng et al., 2015).

2.3.3. Sequential backward selection based optimization

In addition to DE-based optimization, this study introduces a Sequential Backward Selection (SBS) method as a deterministic and low-tuning approach for SPS optimization. While evolutionary algorithms such as DE possess strong global search capability, their practical performance may depend on algorithmic settings and random initialization. By contrast, SBS follows a greedy backward-elimination principle: starting with all candidate sensors, it iteratively removes the sensor whose exclusion least degrades detection performance (quantified by ADT), thereby generating nested subsets adaptable to different budgets. Although SBS does not guarantee a global optimum, it can rapidly provide a stable and engineering-acceptable near-optimal solution when computational budget is limited. The practical viability of SBS is further supported by its application in analogous network optimization problems, such as optimal valve placement in water distribution systems (Wu and Soga, 2023). In the present study, SBS is adapted to the sewer network sensor placement problem as a practical complementary method for rapid engineering screening and budget-wise layout generation.

3. Case studies

Three sewer networks were selected as case studies to evaluate the proposed methodology, as illustrated in Fig. 5. The first case (Fig. 5a) depicts a synthetically designed sewer network with a basic configuration of 3 nodes, 3 pipes, and 1 outfall. This simplified structure was intentionally designed to facilitate a clear demonstration of the evaluation metrics. Figs. 5b and 5c present the topologies of two real-world sewer networks, offering a more realistic validation context.

The first real-world network (Fig. 4b) serves an area of 7 km², comprising 55 km of pipelines (diameters 0.4–1.5 m) that convey an average daily flow of 27,202 m³. Its hydraulic model was simplified to 150 nodes and 149 pipes, with simplification restricted to removal of minor terminal laterals (small-diameter, low-flow branches). Post-optimization analysis indicates that this simplification is unlikely to have a material impact on the sensor placement recommendations, as sensors on low-flow terminal laterals contribute marginally to overall detection performance and are therefore consistently deprioritized in the optimized layouts.

A critical feature, highlighted in Fig. 5b, is the network's significant hydraulic heterogeneity. Its upper region, which comprises only 16.7 % of the service area, contributes roughly 32 % of the daily flow—nearly double the network average—due to higher building density. This uneven flow distribution, typical of urban cores, creates highly variable hydraulic backgrounds that can mask infiltration and inflow signals. This characteristic makes the network an ideal case for investigating how such heterogeneous flow patterns influence SPS optimization, a critical factor that has not been examined in previous studies.

The second real-world network (Fig. 5c) is notably larger and more complex, covering an area of about 15 km² with approximately 60 km of pipelines (diameters ranging from 0.3 to 1.5 m) conveying a daily flow

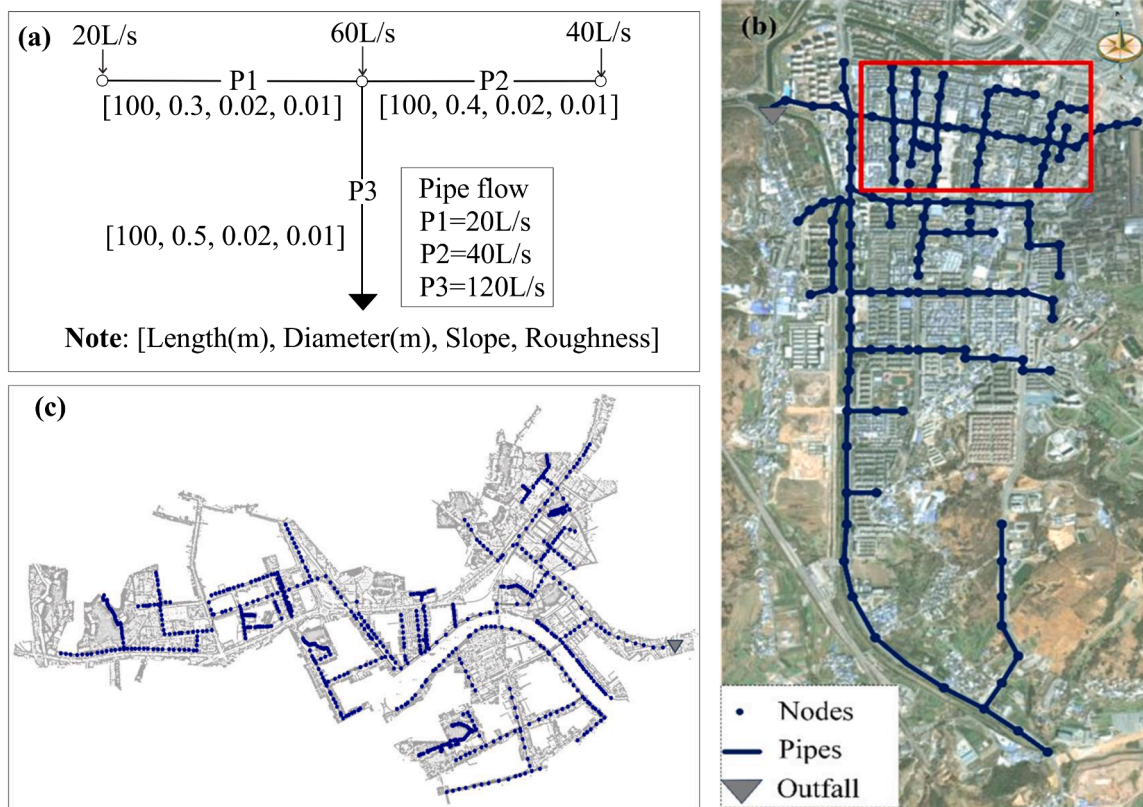


Fig. 5. Overview of One Synthetic and Two Real Sewer Networks.

of about 43,400 m³. Its hydraulic model comprises 820 nodes and 819 pipes. The substantial scale creates a vast solution search space, making the network a challenging testbed for evaluating algorithm performance. Importantly, two benchmark SPSs are available for comparison: an expert-based layout from the local water authority (20 planned flow meters) and an optimization-based SPS generated by Wu et al. (2025) using a deterministic two-stage clustering method. These established benchmarks provide a foundation for rigorously comparing the SPSs generated by our method, enabling a comprehensive assessment of its performance and potential superiority.

4. Results and discussion

4.1. Demonstrating the advantages of DT metric

This section demonstrates the advantages of the proposed DT metric over the traditional EDR metric through a comparative analysis of six SPSs in a three-pipe sewer network, with the results shown in Figs. 6 and 7. Specifically, Figs. 6a–6f present the IPDT and ADT values for each SPS under a 15 % variation tolerance, with Fig. 6g providing a summary of the ADT values for direct visual comparison. Fig. 6h shows the ADT values under a reduced 7.5 % variation tolerance, demonstrating the metric's sensitivity to changes in tolerance levels. For unmonitored pipes, IPDTs are calculated using Eq. (4).

Figs. 7a–7f present the EDR for the same six SPSs, simulating a 12 L/s inflow and infiltration (I&I) scenario with a 15 % variation tolerance. Fig. 7g summarizes the EDR values, and Fig. 7h presents the EDR results under a 7.5 % tolerance to examine the impact of this variation. EDR is computed using binary detection logic with isolated DTs of 3, 6, and 18 L/s. A pipe is highlighted in red if the 12 L/s infiltration event exceeds its assigned threshold, indicating its contribution to event detection.

As illustrated in Fig. 6, the proposed ADT metric effectively distinguishes the detection performance of the six SPSs, assigning them

distinct values of 360 L/s (a), 293 L/s (b), 18 L/s (c), 234 L/s (d), 11.2 L/s (e), and 10.9 L/s (f). In contrast, Fig. 7 reveals that the EDR metric fails in this regard, as four distinct SPSs (a, b, e, f) yield identical EDR values of just 33.3 %. This stark contrast demonstrates the advantages of the proposed DT metric while underscoring the limitation of the EDR.

Notably, the optimal SPSs identified by the proposed ADT metric align well with established engineering principles that prioritize high-flow pipes for enhanced overall detection effectiveness. Indeed, for single-sensor configurations, the best performance occurs with a sensor on the highest-flow pipe (Pipe 3, Fig. 6c; ADT = 18 L/s), while the worst occurs on the lowest-flow pipe (Pipe 1, Fig. 6a; ADT = 360 L/s). Similarly, in dual-sensor layouts, monitoring the two highest-flow pipes (Pipes 2 and 3, Fig. 6f; ADT = 10.9 L/s) outperforms monitoring low-flow pipes (Pipes 1 and 2, Fig. 6d; ADT = 234 L/s). In contrast, the EDR metric produces SPSs that contradict these principles: for a single sensor, placement on the highest-flow pipe is rated worst (Fig. 6c), and for dual sensors, deploying on the two lowest-flow pipes is rated best (Fig. 6d). This stark contradiction indicates that relying on EDR could lead to suboptimal SPSs in practice.

The limitations of the EDR metric stem from its binary assessment method, which determines detectability solely according to whether a threshold is exceeded, without taking into account the magnitude of the exceedance. Additionally, the EDR exhibits high sensitivity to the assumed magnitudes of I&I events and variation tolerance levels. For instance, if the variation tolerance level drops from 15 % to 7.5 %, the EDR values for SPSs (c), (e), and (f) rise to 100 % because all events then become detectable by the sensor in Pipe 3 (Fig. 7h).

In contrast, the proposed IPDT metric overcomes these limitations, as demonstrated by the distinct ADT values assigned to different SPSs in Fig. 6. Moreover, the relative rankings of these ADT values remain consistent across different variation tolerance levels, ensuring consistent performance evaluation. For example, with variation tolerance reduced to 7.5 % (Fig. 6h), the ADT values for the six SPSs are 359 L/s, 291 L/s,

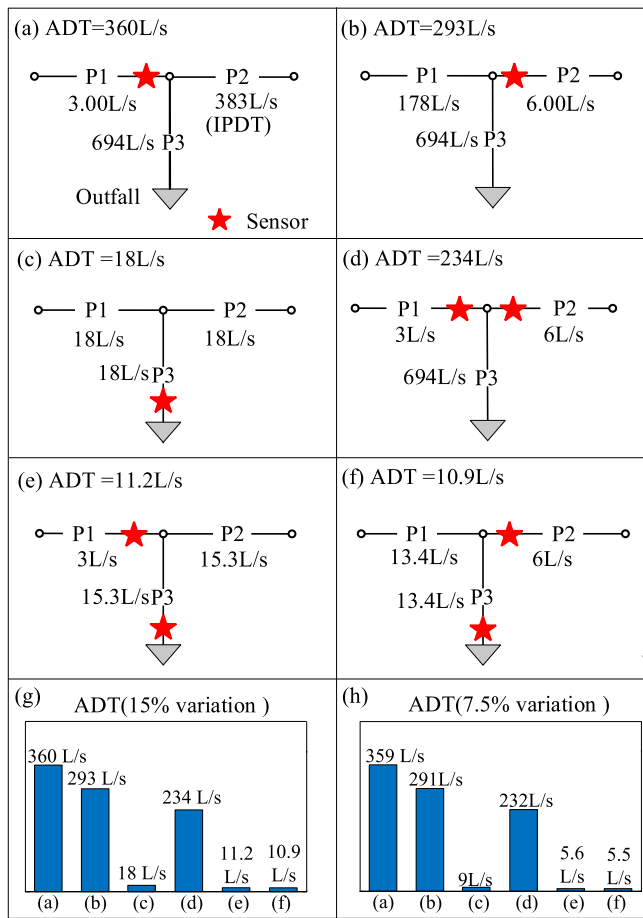


Fig. 6. The Individual Pipe Detection Threshold (IPDT) and Average Detection Threshold (ADT) of six SPSs under 15 % and 7.5 % variation tolerances.

9.0 L/s, 232 L/s, 5.6 L/s, and 5.5 L/s, maintaining the same relative order as in Fig. 6g. This robustness arises because IPDT scales linearly with variation tolerance, enabling consistent relative comparisons even when the exact tolerance is uncertain—a common practical scenario.

4.2. Comparison of layout characteristics between different SPSs

This section compares the layout characteristics of SPSs optimized for ADT minimization with those optimized for DU maximization. The comparison is motivated by a key practical concern: while superior detection sensitivity can be achieved by ADT-minimized SPSs, their field applicability depends critically on achieving a balanced spatial distribution. Therefore, analyzing the underlying layout rationale and mechanism is of paramount importance.

Fig. 8 presents this comparative analysis based on the first real sewer network. Fig. 8a illustrates the spatial distributions of the two 19-sensor SPSs, optimized using the DE algorithm with distinct objectives: minimizing the system-wide ADT versus maximizing DU. Fig. 8b further compares the density distributions of IPDTs, providing insight into the detection performance across the network. Notably, the use of a 19-sensor setup is for methodological demonstration rather than a recommendation for field deployment. This relatively high number was chosen to provide a rich solution space, revealing the inherent layout contrasts between the two optimization objectives.

In terms of sensor distribution, Fig. 8a reveals a high degree of consistency between two strategies. Among the 19 sensors, seven are placed in identical positions, seven are located within one or two pipe segments of each other, and the remaining five, indicated by yellow pentagrams, occupy different positions in the two SPSs. A co-location

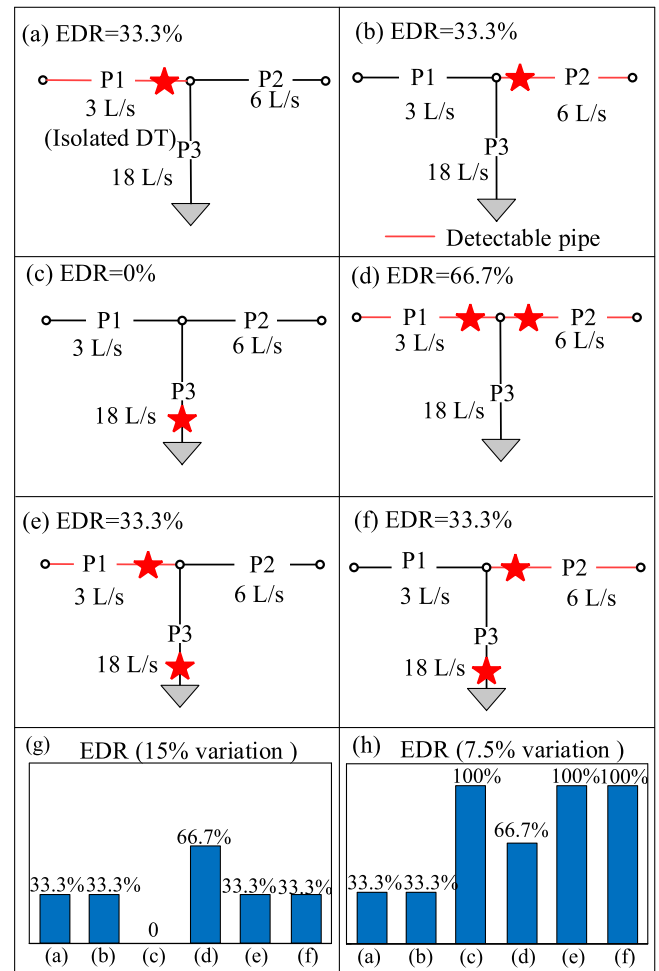


Fig. 7. The Event Detection Reliability (EDR) of six SPSs under 15 % and 7.5 % variation tolerances.

and close-proximity rate of 74 % indicates that the ADT-minimized SPS maintains spatial equity while prioritizing detection sensitivity, thereby affirming its practical feasibility.

Regarding sensor placement adjustments, two notable observations can be made. First, the ADT-minimized SPS adds two sensors in the upper region (now 6, up from 4) and removes one from the lower region (now 2, down from 3), as indicated by purple rectangles in Fig. 8a. This shift is driven by the fact that upper region's higher building density and greater wastewater discharge create more variable hydraulic conditions, requiring more sensors to achieve a low DT. In contrast, the lower region's stability, due to lower density and inflow, allows for fewer sensors without degrading performance.

Second, five sensors (marked by yellow pentagrams) have been strategically relocated to branch pipes with larger flow rates. This placement significantly lowers the DT for these critical high-flow branches. Although sensors were consequently removed from the main pipe, its DT did not rise substantially. This is because applying flow conservation (as illustrated in the previous section) enables effective inference of the main pipe's flow, preventing significant degradation in its detection capability.

In terms of detection performance, sensor placement adjustments obtained using the ADT metric reduced the IPDT by over 2 L/s for 11 longer branch pipes (red markers, Fig. 8a), versus only 2 shorter pipes increased IPDT (blue markers). This result is expected as our method weights pipe length in the objective, as longer pipes are more prone to I&I. Fig. 8b further demonstrates the superior performance of the ADT-minimization SPS, with IPDTs concentrated in the 2–8 L/s range (red

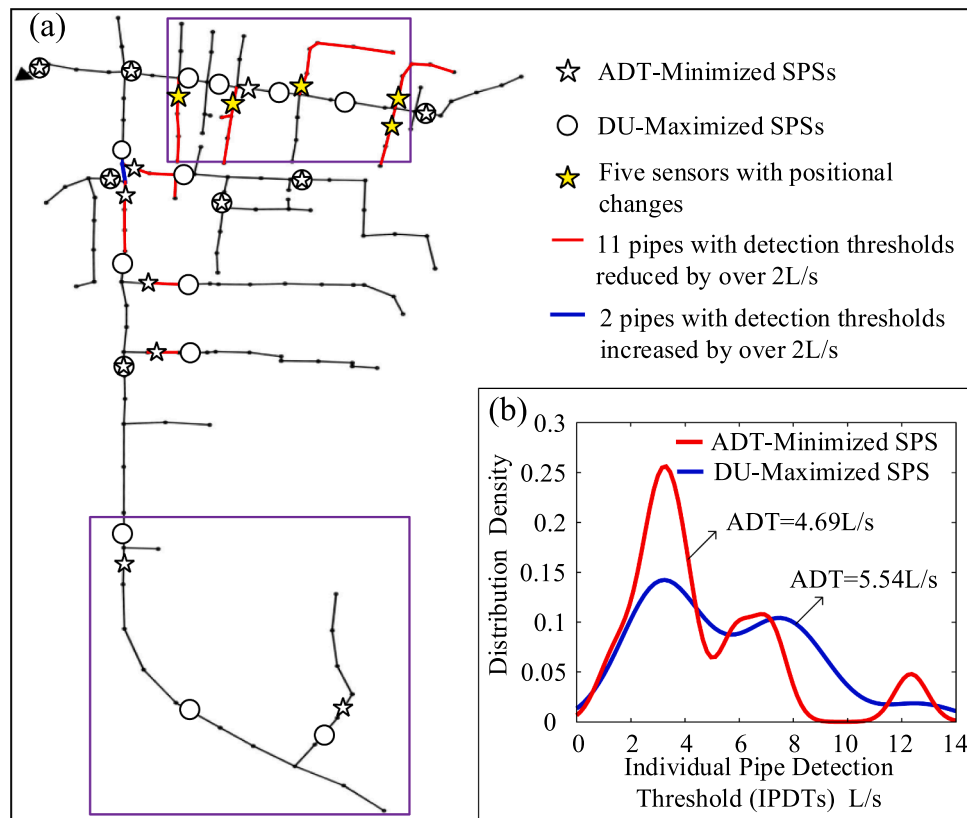


Fig. 8. ADT-minimized vs. DU-maximized SPSs applied to the real network; (a) Layout comparison; (b) IPDTs distribution density.

curve), whereas a significant portion of the DU-maximized results lie between 8 and 10 L/s (blue curve). Consequently, our approach yields a lower ADT of 4.69 L/s—approximately 15 % below the 5.54 L/s ADT of the DU-maximized SPS. This indicates that the proposed method produces superior SPSs capable of detecting potential minor I&I events, in contrast to the DU-maximized method.

In summary, the ADT-minimized SPS maintains broad spatial coverage while allocating a number of sensors in high-drainage zones characterized by greater flow uncertainty, where additional sensors are required to lower DTs. This distribution of sensors enhances detection performance compared to the DU-maximized approach by enabling the identification of smaller I&I events.

4.3. SBS method validation

This section aims to validate the effectiveness of the proposed SBS method. Although SBS offers determinism and computational efficiency, its iterative, greedy nature risks convergence to local minima, potentially resulting in performance significantly inferior to globally optimal solutions obtained via the DE algorithm. Therefore, a comprehensive comparative analysis is essential to benchmark SBS against the DE algorithm. Additionally, the impact of non-uniform drainage patterns on the optimal SPS is investigated.

The experimental setup for this comparison was as follows. The SBS method was initialized with all 149 candidate sensors and iteratively pruned the sensor contributing the least to the detection performance. In contrast, the DE algorithm used a population size of 150 and was run for 500 generations, employing the F and Cr parameters (0.5 and 0.3) recommended by Zheng et al. (2015). Fig. 9 presents the results of the comparison between SBS and DE-based optimization.

From Fig. 9a, the distribution of the 30 sensors (a number chosen for methodological clarity rather than to represent an actual deployment) reveals that the proposed SBS method inherently prioritizes placement

in hydraulically critical regions. Specifically, nearly half of the sensors (14 out of 30) are concentrated in the upper red rectangular area, which occupies <20 % of the drainage area. This targeted distribution stems from the upper region's higher building density and greater wastewater discharge, conditions that lead to increased flow variability. Consequently, a denser sensor deployment is required in this area to maintain low DTs. Therefore, the SBS method demonstrates an inherent adaptability to the network's key hydraulic features. It effectively allocates more monitoring resources to areas of higher flow uncertainty, which are critical for maintaining robust system-wide detection performance.

Fig. 9b demonstrates the sequential removal of 11 sensors from an initial set of 30, highlighting the practical utility of the SBS method. The eliminated sensors are predominantly located in terminal branches (e.g., sensors 2–9 and 11), and their removal has minimal impact on overall detection performance. This is because flow variations in these branches contribute less to downstream uncertainty—a feature systematically captured by the backward-elimination mechanism of SBS, which iteratively identifies and removes the least influential sensors. Moreover, the SBS procedure naturally generates a strict subset hierarchy of sensor layouts, ranging from the full set down to a single sensor. This hierarchy enables clear prioritization for phased deployment and supports cost-effective network expansion, as sensors can be objectively ranked based on their incremental contribution to detection performance.

Notably, Fig. 9c demonstrates strong spatial consistency between the SBS and DE-optimized 19-sensor layouts, as evidenced by 12 co-located sensors and the remaining 7 in close proximity. This spatial similarity is reinforced by the performance data in Fig. 9d, which shows that the ADT gaps between SPSs from the two methods are minimal, with only three exceeding a 5 % difference. Furthermore, for SPSs with >40 sensors, the relative ADT differences become negligible (and are thus omitted from the figure for clarity), underscoring the high degree of convergence between the two optimization approaches.

In conclusion, the SBS method delivers near-optimal SPSs, matching

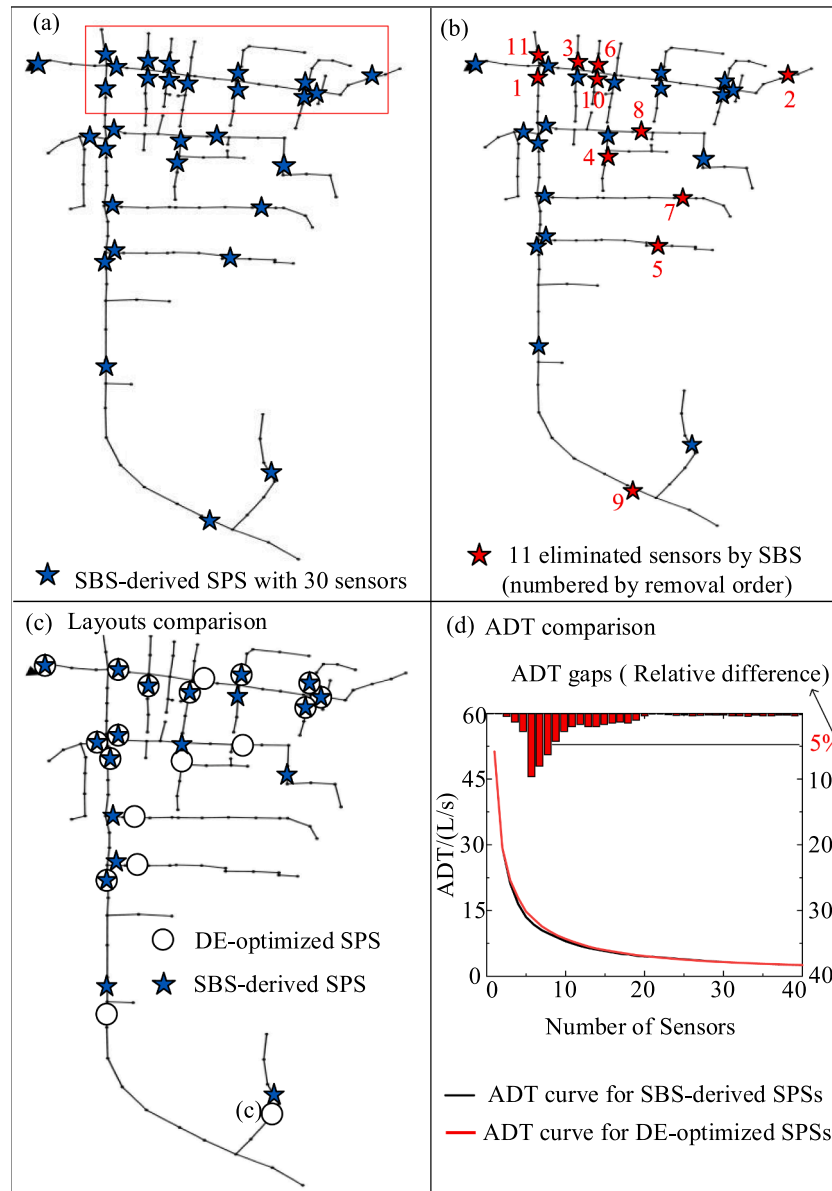


Fig. 9. Comparison of sequential backward selection (SBS) and differential evolution (DE)-based optimization: (a) SBS-derived sensor placement strategy (SPS) with 30 sensors; (b) Progressive sensor reduction from 30 to 19 using SBS; (c) Comparison of 19-sensor layouts between SBS and DE optimization; (d) ADT curves and gaps for both strategies across different sensor counts.

the performance of the DE algorithm. For practical use, a hybrid approach is recommended: first, use the SBS for rapid ADT estimation and cost-benefit analysis; then, use its SPS to seed the DE algorithm's initial population. This ensures fast convergence to a robust, global optimum. As Fig. 10 shows for a 19-sensor layout, DE initialized with an SBS solution converges smoothly to a superior result in under 100 generations, whereas random initialization causes volatile performance. Consequently, seeding DE with a high-quality SPS enables reliable and efficient optimization.

4.4. Validation on the large scale sewer network

This section validates the proposed methodology on the large-scale sewer network through two key comparisons. First, we evaluate the sensitivity gain of the TDFI method relative to the isolated detection approach, wherein each sensor uses a fixed, sensor-specific threshold without exploiting the topological flow relationships, by analyzing IPDT distributions derived from the expert-driven SPS (Fig. 11). Second, we

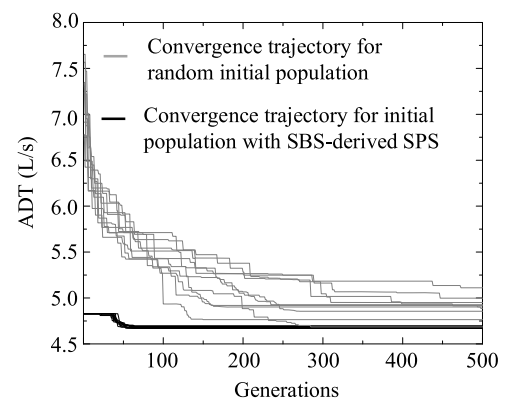


Fig. 10. DE convergence comparison with SBS-initialized versus random population.

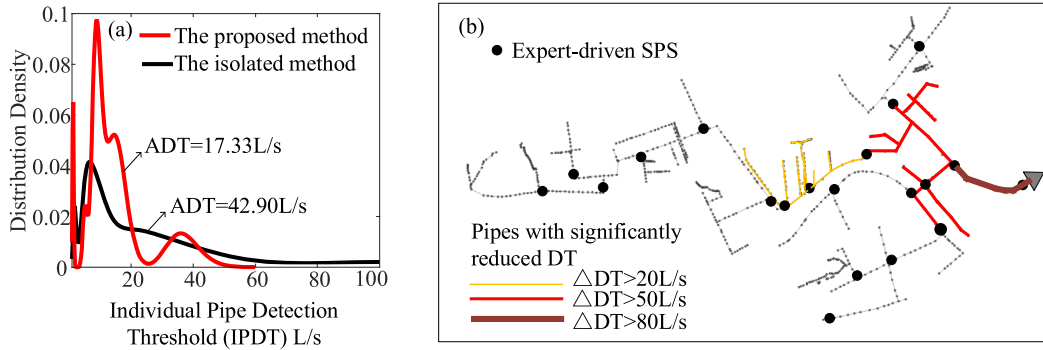


Fig. 11. Comparison of individual Pipe Detection Threshold Distributions between the proposed and isolated methods applied to the expert-driven SPS.

assess the competitiveness of the proposed SBS method by comparing three SPSs: the SBS-derived layout, the DE-optimized layout, and the benchmark from Wu et al. (2025), with a focus on spatial arrangement and IPDT distribution (Fig. 12).

From Fig. 11a, it is evident that the proposed TDFI method achieves a substantially lower and more concentrated distribution of IPDT compared to the isolated method. The ADT of 17.33 L/s for the proposed method is only 40.3 % of the isolated method's 42.90 L/s, confirming a remarkable improvement in detection sensitivity. This improvement stems from the ability of the proposed approach to leverage flow information from upstream sensors, which effectively constrains flow uncertainties. Furthermore, Fig. 11b reveals a distinct spatial pattern in performance gains. The most significant threshold reductions ($\Delta DT > 50$ L/s) consistently occur in downstream trunk sections. This trend arises because the more upstream sensors whose data can be utilized at a downstream sensor, the greater the constraint on flow uncertainty, which in turn allows for a larger reduction in the DT.

Fig. 12a shows that the SBS-derived and DE-optimized SPSs achieve comparable detection performance, and both are superior to the SPS from Wu et al. (2025). This is evidenced by the leftward shift and close alignment of their IPDT distribution curves relative to Wu's in this figure. Quantitatively, the SBS and DE methods yield significantly lower ADT values (10.41 L/s and 10.22 L/s, respectively) versus 14.14 L/s for Wu's SPS.

The comparison of SBS and DE performances is further demonstrated in Fig. 12b, which shows a high degree of spatial overlap: 13 of 20 sensor locations are identical, with the remaining 7 only slightly different. This strong consistency confirms the effectiveness of the SBS method in producing solutions comparable with the DE algorithm in a large-scale sewer network. The superior performance of the proposed SPSs can be attributed to three physics-informed design principles evident in Fig. 12b: (i) elimination of redundant sensors on low-flow branches, as indicated by the purple rectangle in the upper right; (ii) increased sensor density on key high-flow paths, shown in red; and (iii) strategic coverage of previously unmonitored sections of the main trunk, as highlighted by the rectangle near the outfall. Collectively, these principles not only explain the performance gains but also translate into actionable,

interpretable guidance for the design of SPSs in practical applications.

5. Conclusions

Inflow and infiltration (I&I) in sewer networks present significant challenges for wastewater management systems, increasing delivery loads and elevating overflow risks that compromise environmental safety. While sensor-based detection offers promising solutions, conventional approaches relying on fixed, isolated DTs fundamentally limit the detectability of smaller I&I events. This limitation stems from their inability to synthesize information from multiple, correlated sensors, despite the tree-like topology of sewer networks that creates deterministic mass-balance relationships.

This study introduces a novel Topology-derived Flow Inference (TDFI) method to overcome these limitations. The method utilizes the inherent mass-balance constraints of sewer networks, enabling the quantification of pipe-specific minimum detectable I&I flows through the intelligent synthesis of upstream sensor data. In addition, we develop a deterministic Sequential Backward Selection (SBS) method, which constructs hierarchical sensor subsets by iteratively removing the least-contributing sensors. The effectiveness of the proposed framework is evaluated through comparisons with traditional metrics and the DE optimization algorithm on both synthetic and real-world sewer networks.

Based on the results obtained the findings and implications of this study are as follows:

- (i) The proposed DT metric demonstrates superior discriminability and robustness compared to the traditional Event Detection Reliability (EDR) metric. In a simplified three-pipe network, the DT metric effectively distinguished various SPSs through distinct ADT values, while the EDR metric failed to differentiate between different sensor layouts, confirming the DT metric's effectiveness for SPS optimization.
- (ii) The DT-minimized SPS enhances detection sensitivity without compromising spatial coverage. In the first real-world network, this approach reduced ADT by approximately 15 % compared to a

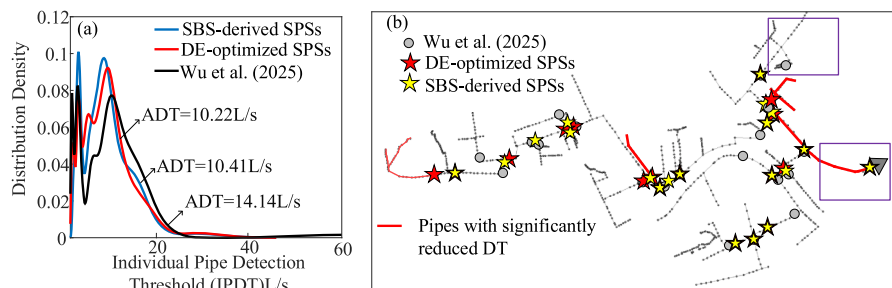


Fig. 12. Comparison of SBS-derived, DE-optimized and Wu et al. (2025)'s SPSs.

DU-maximized SPS. The strategy of concentrating sensors in zones characterized by high building density and significant wastewater discharge has proven essential for improving detection performance, validating the method's ability to balance sensitivity with spatial consistency.

- (iii) The SBS method provides deterministic and computationally efficient solutions as a practical complementary approach to the stochastic DE algorithm. In the tested cases, the layouts generated by SBS showed ADT values close to those obtained by DE, with substantially lower computational effort. Although SBS does not guarantee a global optimum, it can rapidly provide stable near-optimal solutions for practical SPS design (see Fig. 9 and Fig. 12).
- (iv) In the large-network case examined in this study, the proposed methodology improved detection sensitivity relative to the benchmark strategies considered. The SBS-derived placement showed performance close to that of the DE-based solution, indicating the practical usefulness of the framework for SPS design.

While the proposed method offers readily implementable SPS designs for practical use, its performance remains subject to modeling simplifications. These limitations stem primarily from the assumptions of quasi-instantaneous mass conservation under unsteady-flow conditions, common variation tolerance, and zero covariance among aggregated flow components. Accordingly, the present uncertainty formulation should be interpreted as an engineering-oriented approximation for DT-based SPS comparison. To further improve the framework's physical representativeness, future work should focus on: (i) quantifying the impact of unsteady flow on mass-balance validity, (ii) developing dynamic correction mechanisms for flow travel-time effects, (iii) establishing field-data-driven calibration protocols to refine variation tolerance levels, and (iv) conducting broader field-based validation of the uncertainty formulation using real-network observations. Such efforts would help bridge the gap between the idealized inference model and real-world network dynamics while preserving the core optimization framework.

Data availability

The data and codes are available at <https://doi.org/10.5281/zenodo.18872145>.

CRediT authorship contribution statement

Haipai Wang: Visualization, Investigation, Formal analysis, Data curation. **Kun Du:** Writing – original draft, Validation, Project administration, Methodology, Formal analysis, Conceptualization. **Feifei Zheng:** Writing – review & editing, Supervision, Project administration. **Renli Liang:** Resources, Project administration. **Muhua Lu:** Resources, Project administration. **Zoran Kapelan:** Writing – review & editing, Supervision. **Dragan Savic:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Alfonso, L., Lobbrecht, A., Price, R., 2010. Optimization of water level monitoring network in polder systems using information theory. *Water. Resour. Res.* 46, W12553. <https://doi.org/10.1029/2009WR008953>.
- Banik, B.K., Alfonso, L., Di Cristo, C., Leopardi, A., Mynett, A., 2017. Evaluation of different formulations to optimally locate sensors in sewer systems. *J. Water Resour. Plan. Manag.* 143 (7), 04017026. [https://doi.org/10.1061/\(ASCE\)JWR.1943-5452.0000778](https://doi.org/10.1061/(ASCE)JWR.1943-5452.0000778).
- Du, K., Yu, J., Zheng, F., Xu, W., Savic, D., Kapelan, Z., 2024. A robust multi-objective pressure sensor placement method for burst detection in water distribution systems. *Water. Resour. Res.* 60, e2024WR037258. <https://doi.org/10.1029/2024WR037258>.
- Ge, J., Li, J., Qiu, R., Shi, T., Zhang, C., Huang, Z., Yuan, Z., 2024. A data-driven method for estimating sewer inflow and infiltration based on temperature and conductivity monitoring. *Water. Res.* 261, 122002. <https://doi.org/10.1016/j.watres.2024.122002>.
- Jia, Y., Zheng, F., Maier, H.R., Ostfeld, A., Creaco, E., Savic, D., Langeveld, J., Kapelan, Z., 2021a. Water quality modeling in sewer networks: review and future research directions. *Water. Res.* 202, 117419. <https://doi.org/10.1016/j.watres.2021.117419>.
- Jia, Y., Zheng, F., Zhang, Q., Duan, H.-F., Savic, D., Kapelan, Z., 2021b. Foul sewer model development using geotagged information and smart water meter data. *Water. Res.* 204, 117594. <https://doi.org/10.1016/j.watres.2021.117594>.
- Krstanovic, P.F., Singh, V.P., 1992. Evaluation of rainfall networks using entropy: I. Theoretical development. *Water Resour. Manag.* 6 (4), 279–293. <https://doi.org/10.1007/BF00872281>.
- Lee, J.H., 2013. Determination of optimal water quality monitoring points in sewer systems using entropy theory. *Entropy* 15 (9), 3419–3434. <https://doi.org/10.3390/e15093419>.
- Li, J., Li, X., Liu, H., Gao, L., Wang, W., Wang, Z., Zhou, T., Wang, Q., 2023. Climate change impacts on wastewater infrastructure: a systematic review and typological adaptation strategy. *Water. Res.* 242, 120282. <https://doi.org/10.1016/j.watres.2023.120282>.
- Ostfeld, A., Salomons, E., 2004. Optimal layout of early warning detection stations for water distribution systems security. *J. Water Resour. Plan. Manag.* 130 (5), 377–385. [https://doi.org/10.1061/\(ASCE\)0733-9496\(2004\)130:5\(377\)](https://doi.org/10.1061/(ASCE)0733-9496(2004)130:5(377)).
- Qi, Z., Zheng, F., Guo, D., Maier, H.R., Zhang, T., Yu, T., Shao, Y., 2018. Better understanding of the capacity of pressure sensor systems to detect pipe burst within water distribution networks. *J. Water Resour. Plan. Manag.* 144 (7), 04018035. [https://doi.org/10.1061/\(ASCE\)JWR.1943-5452.0000957](https://doi.org/10.1061/(ASCE)JWR.1943-5452.0000957).
- Tatiparthi, S.R., Whittaker, C.N., Zhuang, W.Q., 2025. Real-time detection of sewer water levels and blockages using UHF-RFID sensors. *Water. Res.* 278, 123380. <https://doi.org/10.1016/j.watres.2025.123380>.
- Taylor, B.N., Kuyatt, C.E., 1994. Guidelines for Evaluating and Expressing the Uncertainty of NIST Measurement Results. NIST Technical Note 1297. National Institute of Standards and Technology, Gaithersburg, MD. <https://doi.org/10.6028/NIST.TN.1297>.
- Wu, R., Soga, K., 2023. Isolation valve placement strategy for resilience improvement of water distribution systems. *J. Water Resour. Plan. Manag.* 149 (8), 04023040. <https://doi.org/10.1061/JWRMD5.WRENG-5933>.
- Wu, Y., Zheng, F., Yang, Y., Zhang, K., Du, K., Duan, H., Savic, D., Kapelan, Z., 2025. Enhancing inflow and infiltration detection in urban sewer networks with a new deterministic sensor placement method. *Water. Res.* 268, 122591. <https://doi.org/10.1016/j.watres.2024.122591>.
- Yazdi, J., 2018. Water quality monitoring network design for urban drainage systems, an entropy method. *Urban Water J.* 15 (3), 227–233. <https://doi.org/10.1080/1573062X.2018.1424215>.
- Zeydinejad, N., Javadi, A.A., Webber, J.L., 2024. Global perspectives on groundwater infiltration to sewer networks: a threat to urban sustainability. *Water. Res.* 262, 122098. <https://doi.org/10.1016/j.watres.2024.122098>.
- Zhang, Q., Zheng, F., Jia, Y., Savić, D., Kapelan, Z., 2021. Real-time foul sewer hydraulic modelling driven by water consumption data from water distribution systems. *Water. Res.* 188, 116544. <https://doi.org/10.1016/j.watres.2020.116544>.
- Zheng, F., Zecchin, A.C., Simpson, A.R., 2015. Investigating the run-time searching behavior of the differential evolution algorithm applied to water distribution system optimization. *Environ. Model. Softw.* 69 (1), 292–307. <https://doi.org/10.1016/j.envsoft.2014.09.022>.